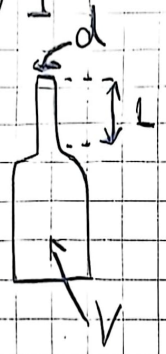
AKU - Akustik im Haushalt 07.03.26

Jonas Müller

TVI:



(Volumenmessung durch Füllen mit Wasser und Wiegen)

$d = (17{,}8 \pm 0{,}05)\,mm$ (Messschieber)

~~$l = (79 \pm 5)\,mm$~~

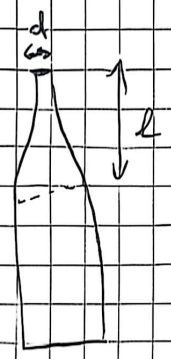
$l = (81 \pm 3)\,mm$ (Messschieber)

⇒ Fehler da Halsende nicht exakt definiert ist.

$$V = \frac{m_{voll}}{\rho_W} = \frac{771\,g}{1000\,\frac{g}{L}} = (0{,}771 \pm 0{,}001)\,L \qquad m_F = 492\,g$$

m_W	f
$73\,g \pm 1\,g$	~~118,4~~ $(119{,}0 \pm 1{,}5)\,Hz$
$(149 \pm 1)\,g$	~~(125,0 ± 1,5) Hz~~ $(125{,}0 \pm 1{,}5)\,Hz$
$(228 \pm 1)\,g$	$(135{,}0 \pm 1{,}5)\,Hz$
$(304 \pm 1)\,g$	$(145{,}0 \pm 1{,}5)\,Hz$
$(377 \pm 1)\,g$	$(160{,}0 \pm 1{,}5)\,Hz$
$(452 \pm 1)\,g$	$(178{,}5 \pm 1{,}5)\,Hz$
$(522 \pm 1)\,g$	$(200{,}5 \pm 1{,}5)\,Hz$
$(602 \pm 1)\,g$	$(248{,}0 \pm 1{,}5)\,Hz$

m_W	f
$(677 \pm 1)\,g$	$(351{,}0 \pm 1{,}5)\,Hz$
$(753 \pm 1)\,g$	$(1260 \pm 10)\,Hz$
$0\,g$	$(112{,}0 \pm 1{,}5)\,Hz$



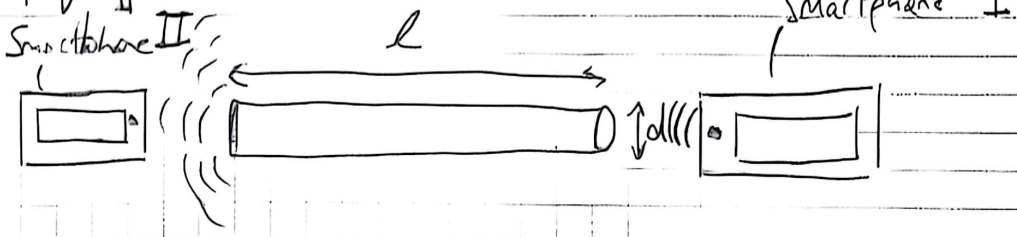
$d = (17{,}03 \pm 0{,}05)\,mm$

$l = (130 \pm 5)\,mm$

$m_{ges} = \frac{m_{voll}}{\rho_{Wasser}} = \frac{517\,g \pm 1\,g}{1000\,\frac{g}{l}} = (0{,}517 \pm 0{,}001)\,kg$

m_W	f
0 g	(155,0 ± 1,5) Hz
(49 ± 1) g	(161,5 ± 1,5) Hz
(101 ± 1) g	(171,0 ± 1,5) Hz
(151 ± 1) g	(183,0 ± 1,5) Hz
(198 ± 1) g	(195,0 ± 1,5) Hz
(254 ± 1) g	(217,0 ± 1,5) Hz
(299 ± 1) g	(240,0 ± 1,5) Hz
(355 ± 1) g	(276,0 ± 1,5) Hz
(403 ± 1) g	(339,5 ± 1,5) Hz
(454 ± 1) g	(463,0 ± 3) Hz

T V II



$l = (28{,}4 \pm 0{,}1)\,cm$ (Meterstab)

$d = (47{,}0 \pm 0{,}2)\,mm$ (Messschieber)